Homework 3

12.4 Consider the following data and parity-block arrangement on four disks:

Disk 1	Disk 2	Disk 3	Disk 4
B_1	B_2	B_3	B_4
P_1	B_5	B_6	B_7
B_8	P_2	B_9	B_{10}
⋮	⋮	⋮	⋮

The B_is represent data blocks; the P_is represent parity blocks. Parity block P_i is the parity block for data blocks B_{4i-3} to B_{4i}. What, if any, problem might this arrangement present?

since the parity block and B_{4i-3} are on the same disk, if the disk fails, rebuilding B_{4i-3} is impossible since the data and parity block are both lost

12.5 A database administrator can choose how many disks are organized into a single RAID 5 array. What are the trade-offs between having fewer disks versus more disks, in terms of cost, reliability, performance during failure, and performance during rebuild?

In a RAID 5 configuration, the trade-offs between having fewer disks versus more disks can include:

Cost: Having fewer disks generally means a lower upfront cost for purchasing the disks and installing the RAID configuration. However, having fewer disks also means less capacity for storing data, which may require more frequent upgrades or additional storage solutions in the future.

Reliability: Having more disks in a RAID 5 configuration can improve reliability by providing more redundancy in case of disk failure. However, having more disks also means that there is a higher likelihood of experiencing a disk failure due to the increased number of disks, which could potentially impact the reliability of the system.

Performance during failure: In the event of a disk failure, a RAID 5 configuration with fewer disks may experience a greater performance impact compared to a configuration with more disks due to the reduced capacity for storing data.

Performance during rebuild: After a disk failure, the RAID 5 configuration will need to rebuild the data on a replacement disk. This process can impact performance, and a RAID 5 configuration with more disks may experience a greater performance impact during the rebuild process due to the increased overhead for calculating and storing the parity information

12.6 A power failure that occurs while a disk block is being written could result in the block being only partially written. Assume that partially written blocks can be detected. An atomic block write is one where either the disk block is fully written or nothing is written (i.e., there are no partial writes). Suggest schemes for getting the effect of atomic block writes with the following RAID schemes. Your schemes should involve work on recovery from failure.

a. RAID level 1 (mirroring)

b. RAID level 5 (block interleaved, distributed parity)

a. RAID level 1 (mirroring)

One way to achieve the effect of atomic block writes in a RAID 1 configuration is to write the data to one of the mirrors, and then simultaneously write the data to the other mirror. If a power failure occurs while the data is being written, the partially written data will be detected and the write operation can be rolled back, ensuring that the data is either fully written or not written at all.

b. RAID level 5 (block interleaved, distributed parity)

One way to achieve the effect of atomic block writes in a RAID 5 configuration is to write the data to one of the disks in the RAID group, and then simultaneously write the data to the remaining disks in the group along with the parity information. If a power failure occurs while the data is being written, the partially written data will be detected and the write operation can be rolled back, ensuring that the data is either fully written or not written at all. Alternatively, the data can be written to a cache or buffer and then committed to the disks in the RAID group in an atomic operation, to achieve the same effect.

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